॥वसुधैव कुटुम्बकम्॥

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Lab: Data Visualization Lab

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Aim: To develop a single-page sales analysis dashboard applying data visualization best practices

Datasets Used:

1. AdventureWorks Sales Data 2022
2. AdventureWorks Calendar Lookup
3. AdventureWorks Customer Lookup
4. AdventureWorks Product Categories Lookup
5. AdventureWorks Product Lookup
6. AdventureWorks Product Subcategories Lookup
7. AdventureWorks Territory Lookup

Additional tables created during Power Query exercises:

8. Student
9. JanuarySales
10. FebruarySales
11. ProductID

These additional tables were created using **Enter Data** specifically for demonstrating Pivot, Unpivot, Append and Merge operations.

Objectives and Implementation:

The practical contains two parts:

1. Power Query operations
2. Sales Analysis dashboard

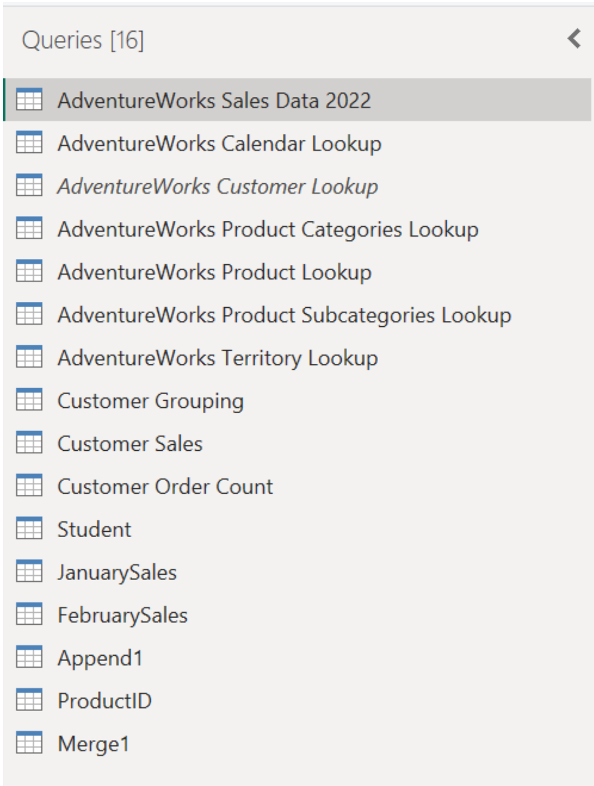
The practical specifically includes Grouping, Pivoting/Unpivoting, Append and Merge operations before the dashboard.

I. Importing the Dataset

The following queries were loaded:

- Sales Data 2022
- Calendar Lookup

- Customer Lookup
- Product Categories Lookup
- Product Lookup
- Product Subcategories Lookup
- Territory Lookup



II. Grouping

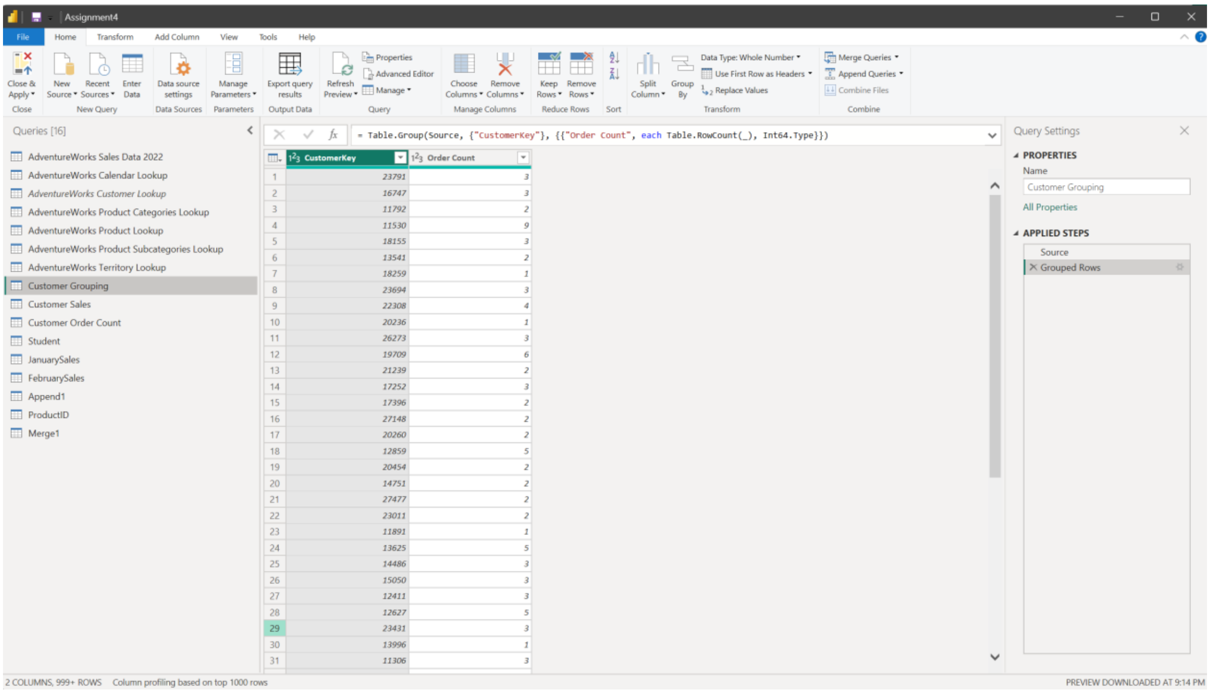
The practical askes 3 questions:

1. How many distinct customers are there?
2. What is the average sales per customer?
3. Which customer has placed the most orders?

For all three I used basic Group By, because each operation required only one field/column and one aggregation

1. Number of Distinct Customers

- Right-clicked **AdventureWorks Sales Data 2022**.
- Selected **Reference**.
- Created a new query named **Customer Grouping**.
- Selected **Home → Group By**.
- Used **Basic Group By**.
- Selected:
  - **Group by:** CustomerKey
  - **New column name:** Order Count
  - **Operation:** Count Rows
- Clicked **OK**.
- Each row represented a unique customer.
- The number of rows in the resulting grouped table represented the number of distinct customers.

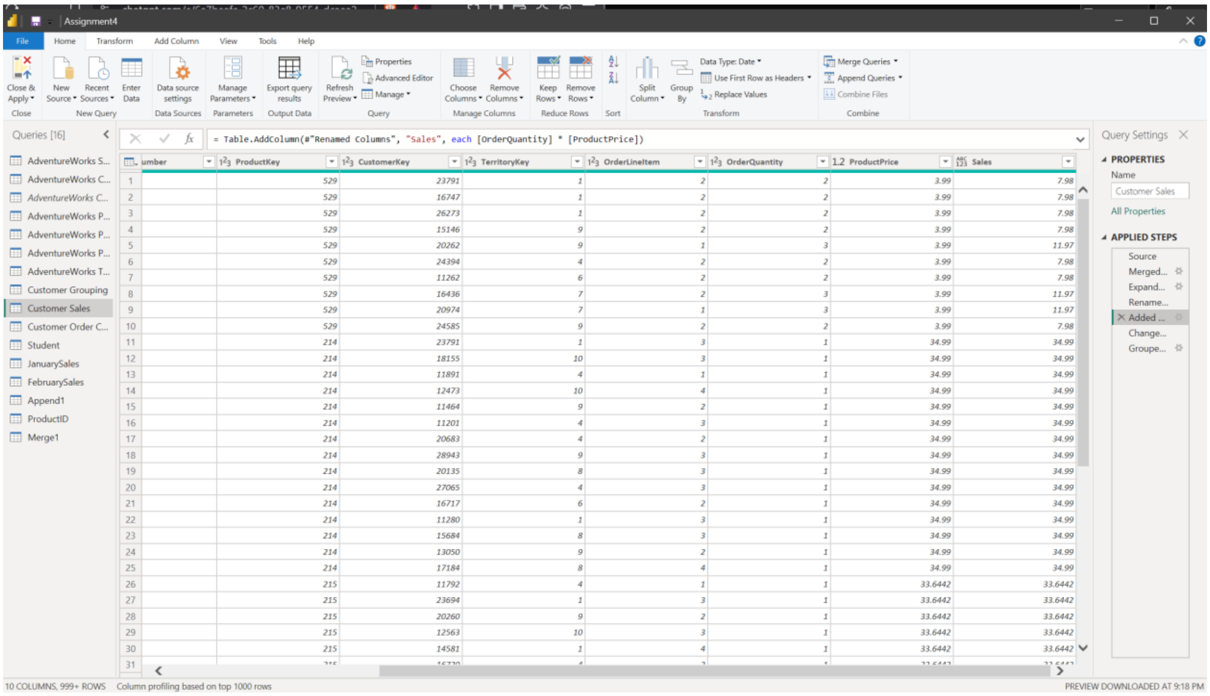


2. Average Sales per Customer

Since the AdventureWorks Sales table did not directly contain a Sales column, sales had to be derived using product information.

- Created a **Reference** of AdventureWorks Sales Data 2022.
- Renamed the query to **Customer Sales**.
- Used **Home → Merge Queries**.
- Merged Sales Data 2022 with **Product Lookup**.
- Used ProductKey as the common column.
- Used a **Left Outer** join.
- Expanded the merged table.
- Selected only ProductPrice.
- Created a Custom Column named **Sales**.
- Used the formula: [OrderQuantity] * [ProductPrice]
- Changed the Sales column to **Decimal Number**.
- Used **Group By → Basic**.
- Grouped by CustomerKey.
- Selected:
  - **New column:** Average Sales
  - **Operation:** Average
  - **Column:** Sales

The sales value was calculated at the order-line level by multiplying the quantity ordered by the product price. Grouping these values by customer allowed the average sales value to be calculated for each customer.


Assignment4
= Table.AddColumn(#"Renamed Columns", "Sales", each [OrderQuantity] * [ProductPrice])
Query Settings
PROPERTIES
Name
Customer Sales
APPLIED STEPS
10 COLUMNS, 999+ ROWS Column profiling based on top 1000 rows
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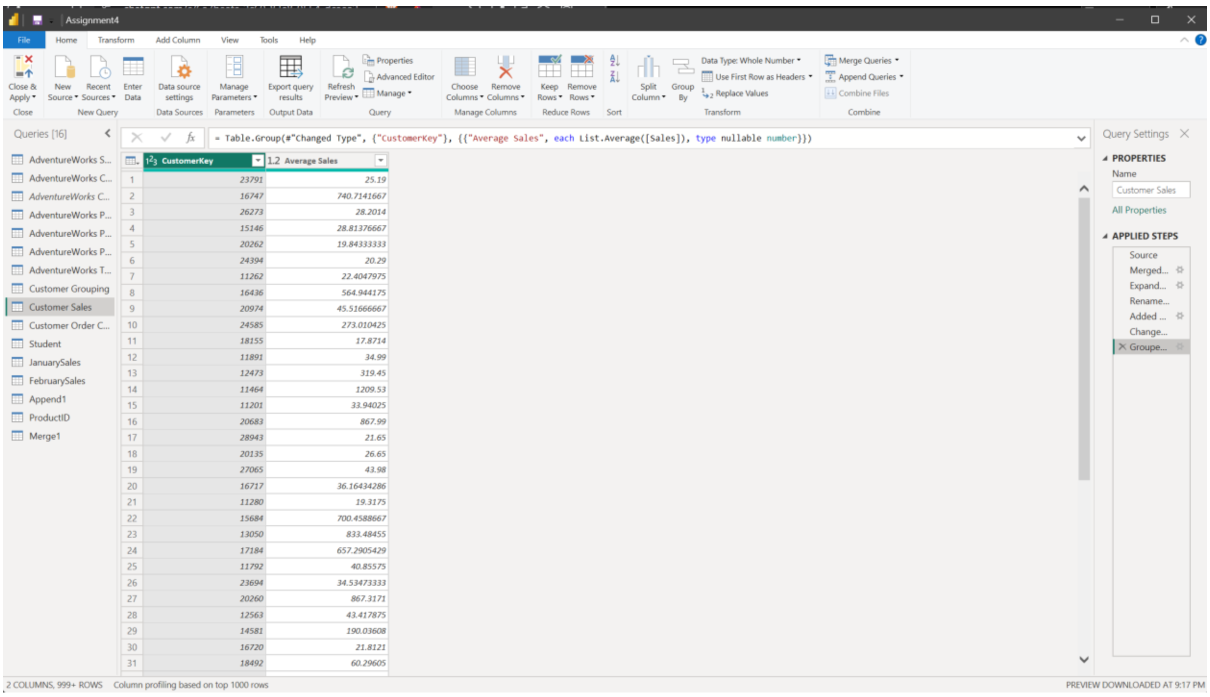
Assignment4
= Table.Group(#"Changed Type", {"CustomerKey"}, {{"Average Sales", each List.Average([Sales]), type nullable number}})
Query Settings
PROPERTIES
Name
Customer Sales
APPLIED STEPS
2 COLUMNS, 999+ ROWS Column profiling based on top 1000 rows
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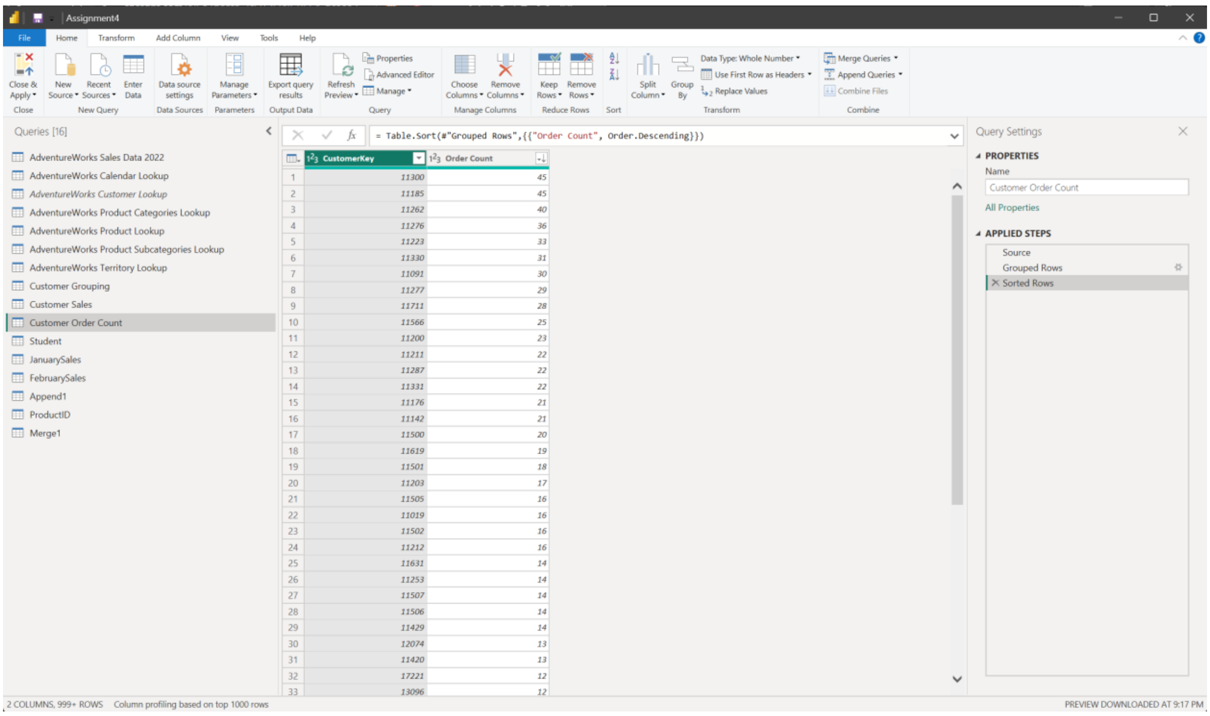
3. Customer with the Most Orders

The practical specifies grouping by customer and counting orders.

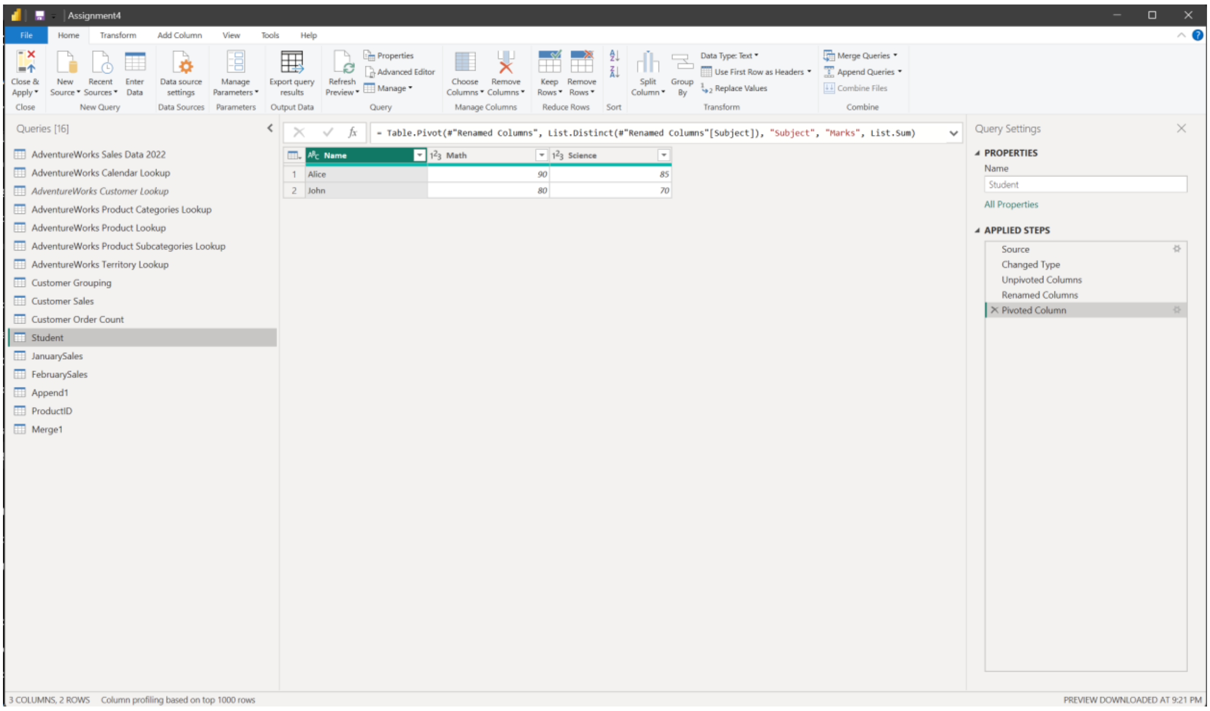
- Created another **Reference** of AdventureWorks Sales Data 2022.
- Renamed it **Customer Order Count**.
- Selected **Home → Group By**.
- Used **Basic Group By**.
- Selected:
  - ➢ **Group by:** CustomerKey
  - ➢ **New column:** Order Count
  - ➢ **Operation:** Count Rows
- Clicked **OK**.
- Sorted Order Count in descending order.
- The customer at the top had the highest order count.



III. Pivoting and Unpivoting

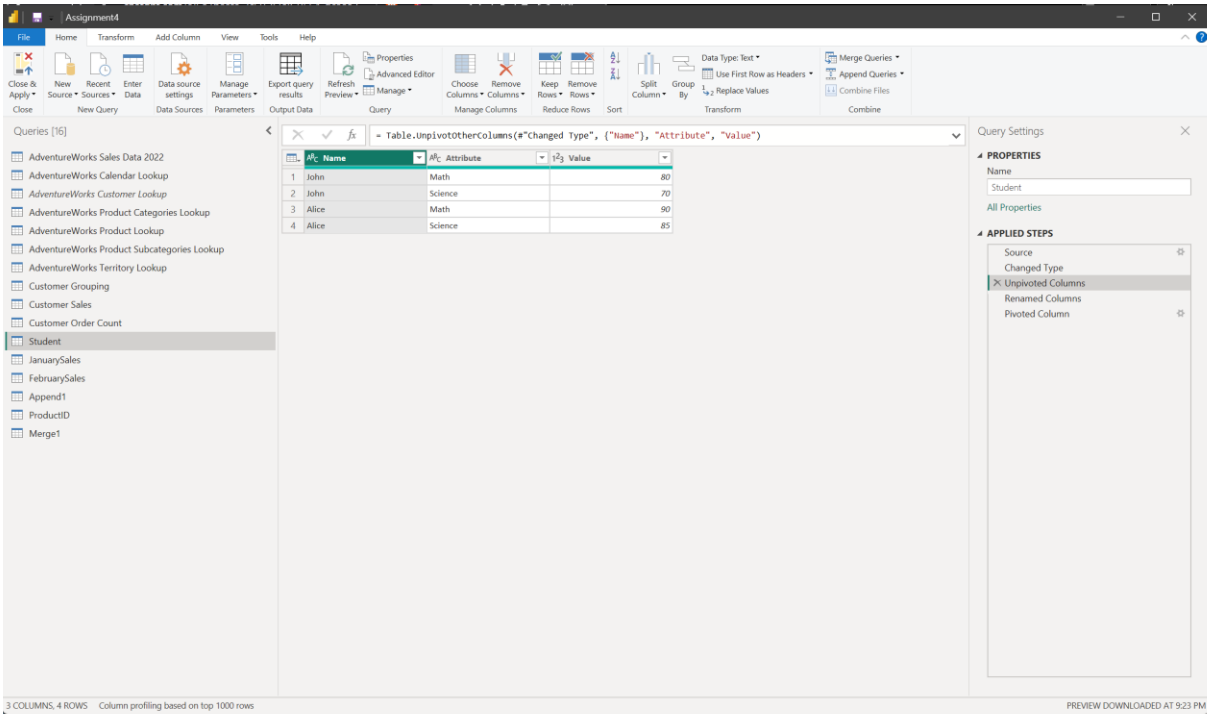
The practical asks us to create a Student table, unpivot Math and Science, rename the resulting columns, and then pivot the table back.

Created a Student table using Home-> Enter Data



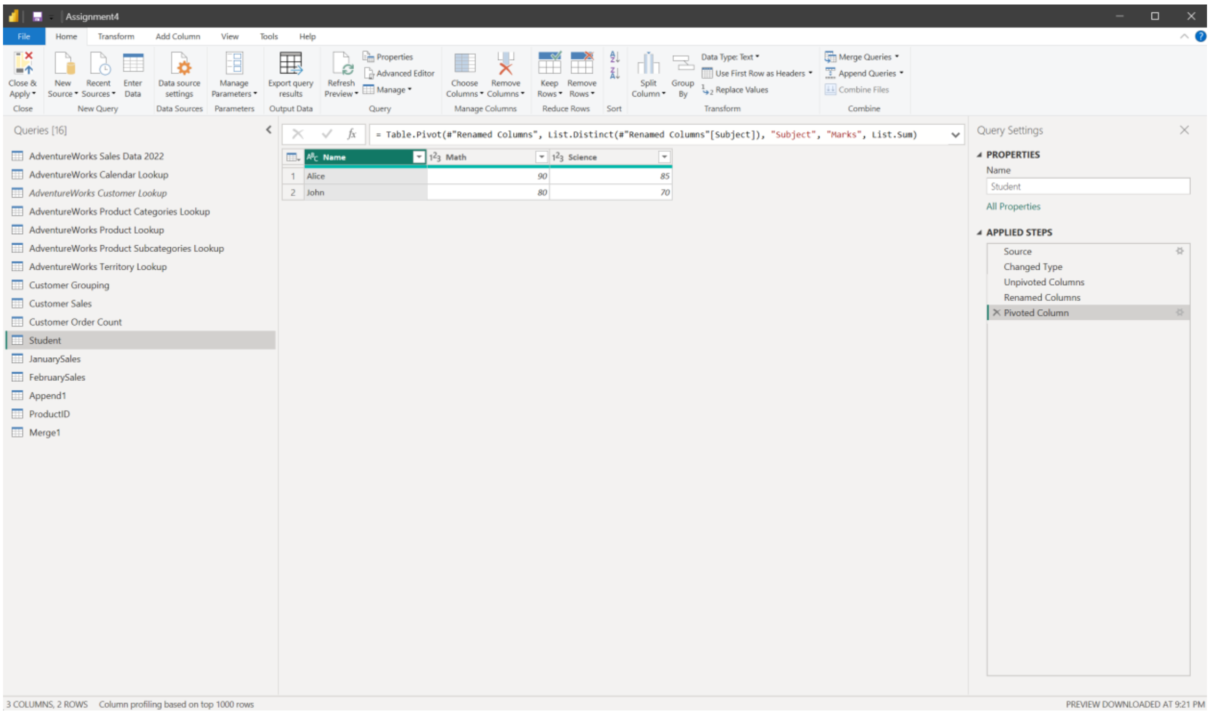
1. Unpivoting

- Selected the **Math** and **Science** columns.
- Went to **Transform** → **Unpivot Columns**.
- The columns were converted into rows.
- The generated columns were:
  - Attribute
  - Value
- Renamed:
  - Attribute → Subject
  - Value → Marks



2. Pivoting

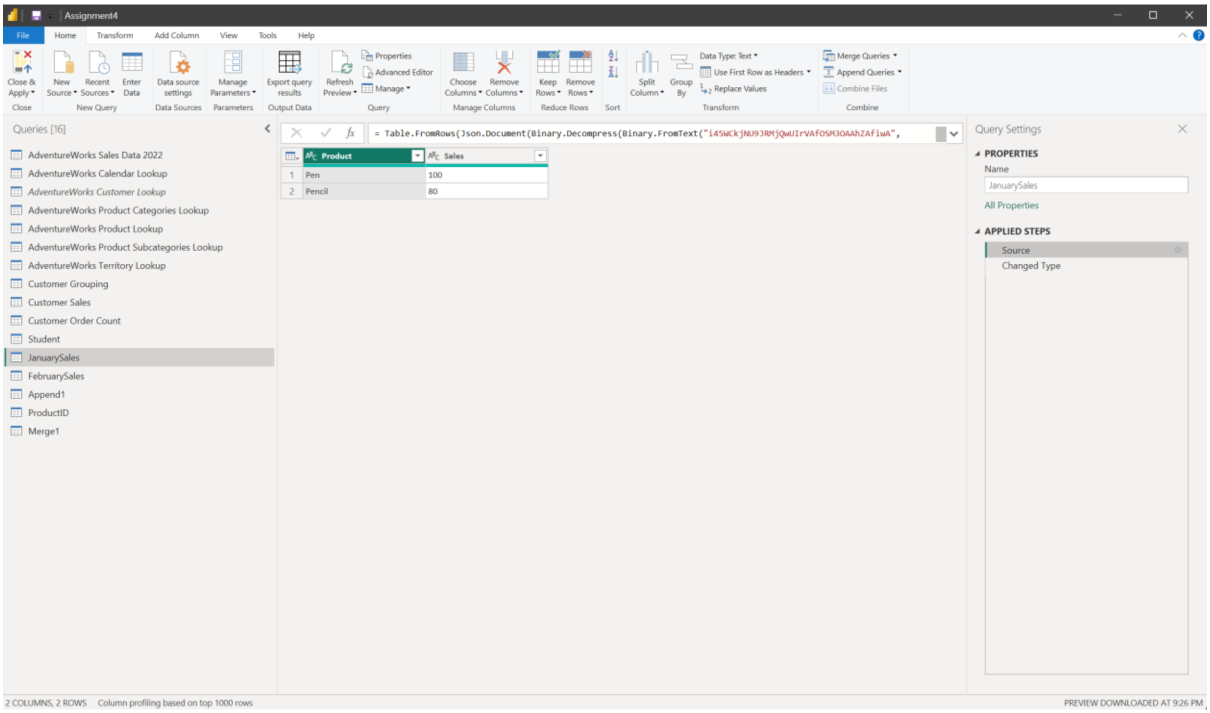
- Selected the `Subject` column.
- Selected **Transform → Pivot Column**.
- Used `Subject` as the column to pivot.
- Selected `Marks` as the **Values Column**.
- Applied the transformation.

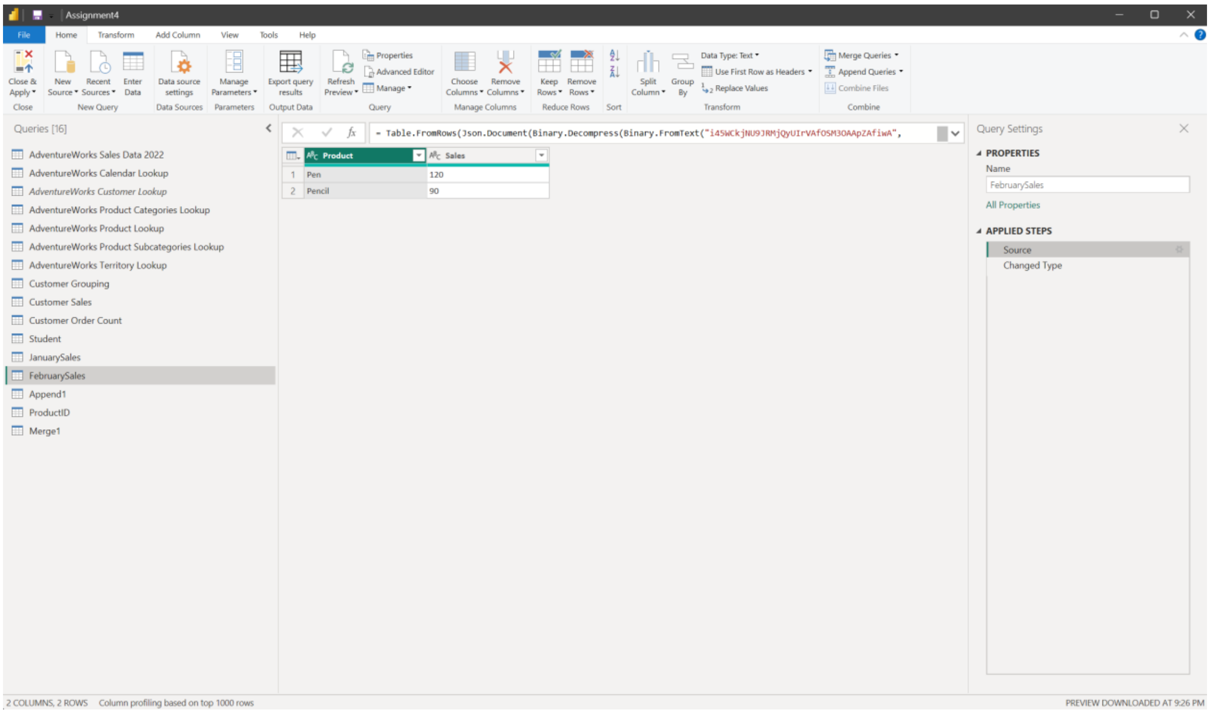


IV. Append Queries

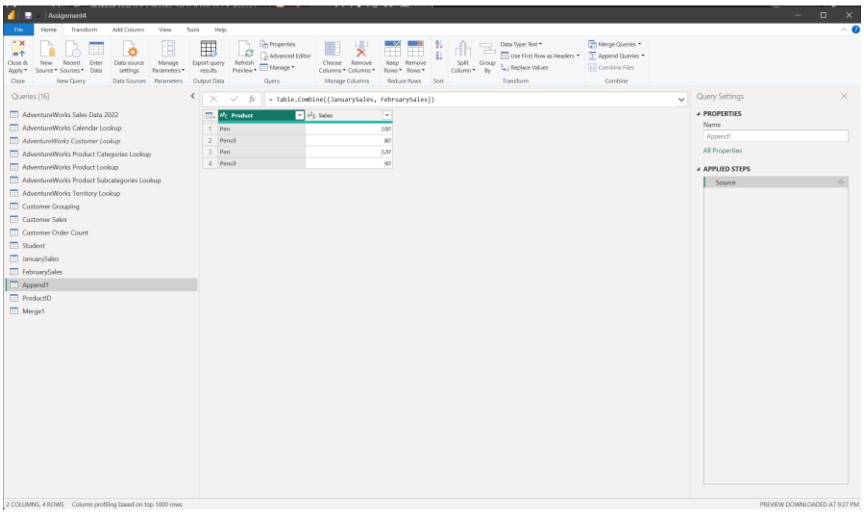
The practical demonstrates Append using JanuarySales and FebruarySales.

JanuarySales and FebruarySales were created.





- Opened Power Query Editor.
- Selected:
  Home → Append Queries → Append Queries as New
- Selected:
  - First table: JanuarySales
  - Second table: FebruarySales
- Clicked **OK**.
- The resulting query was named **Append1**.

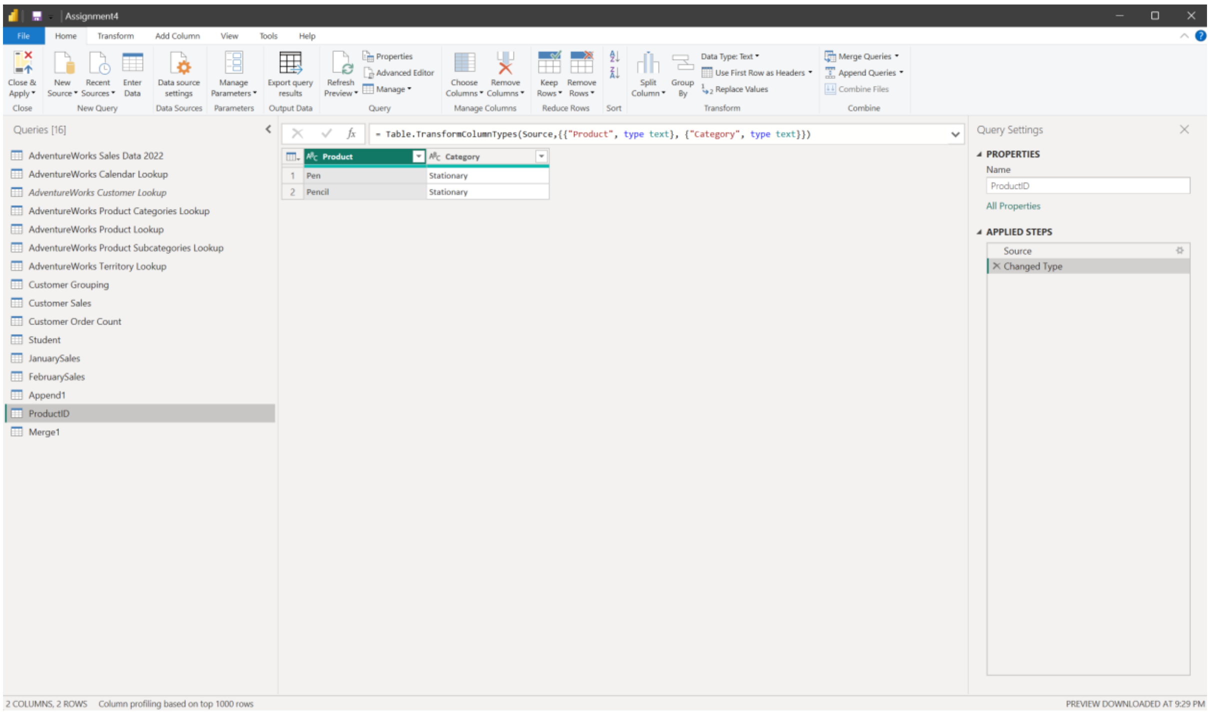


Append is used to **combine rows from two tables vertically** when the tables have the same or compatible column structure.

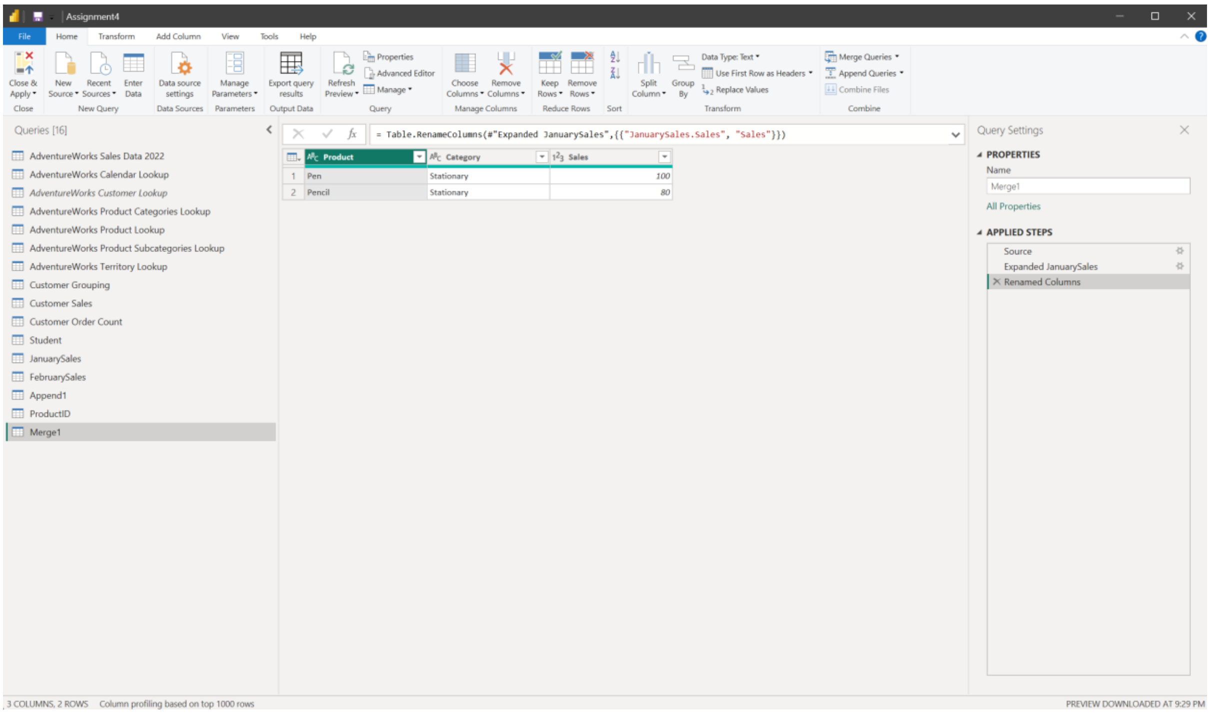
V. Merge Queries

The practical asks for a ProductID table where Product is the common column with JanuarySales, followed by selecting a join kind and expanding the merged table.

- Created a new table using **Enter Data**.
- Named it **ProductID**.



- Selected ProductID.
- Selected **Home → Merge Queries as New**.
- Selected JanuarySales as the second table.
- Selected Product as the common column in both tables.
- Used a **Left Outer** join.
- Clicked **OK**.
- Expanded the merged table.
- Selected the Sales column.



Merge is used to **combine columns from related tables using a common key/field**. Unlike Append, which stacks rows, Merge combines information horizontally.

VI. Sales Analysis Dashboard

After completing the Power Query exercises, the transformed data was loaded into Power BI and a single-page Sales Analysis Dashboard was created.

The practical requires the dashboard to contain four KPI cards, four analytical visuals, a customer-wise table and three slicers.

1. KPI Cards

Four Card visuals were created:

- **Total Sales** – displays overall sales.
- **Total Profit** – displays overall profit.
- **Total Orders** – displays the total number of orders.
- **Total Customers** – displays the number of unique customers.

2. Sales by Category

A **Clustered Column Chart** was created using CategoryName and Total Sales to compare sales across product categories.

3. Monthly Sales Trend

A **Line Chart** was created using Month Name and Total Sales to show monthly sales trends. Month Name and Month Number columns were created to display the months in chronological order.

4. Sales by Region

A **Donut Chart** was created using Region and Total Sales to show the contribution of different regions to overall sales.

5. Top 10 Product by Sales

A **Bar Chart** was created using ProductName and Total Sales. A **Top 10 filter** was applied to display the ten highest-selling products.

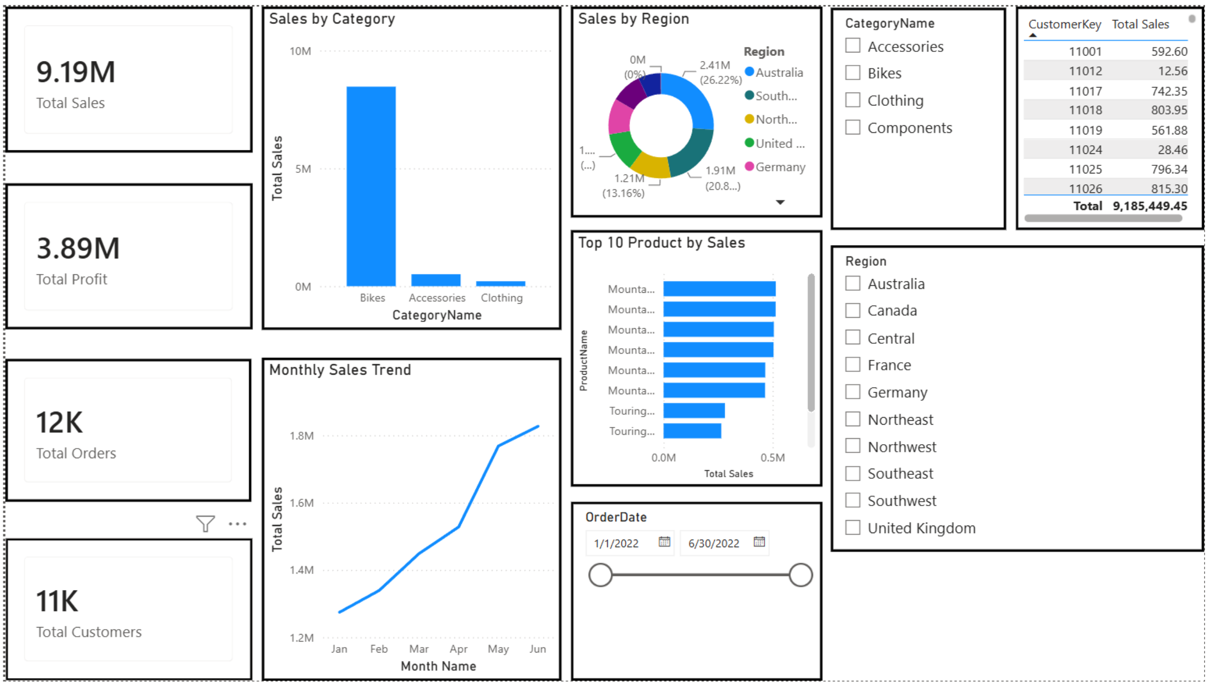
6. Customer-wise Sales

A **Table** was created using CustomerKey and Total Sales to display sales for individual customers.

7. Slicers

Three interactive slicers were added:

- **Order Date** – filters the dashboard by date.
- **Region** – filters sales by region.
- **Category** – filters sales by product category.



Conclusion

The experiment successfully demonstrated the use of **Microsoft Power BI** for transforming sales data and creating an interactive Sales Analysis Dashboard. Power Query was used for grouping, pivoting, unpivoting, appending and merging data, while cards, charts, tables and slicers were used to analyze sales performance.

The final dashboard provides insights into **sales, profit, orders, customers, categories, regions, products and monthly trends**, while interactive slicers allow users to filter the information according to their requirements.